Analysis of a Doubly Spread Channel

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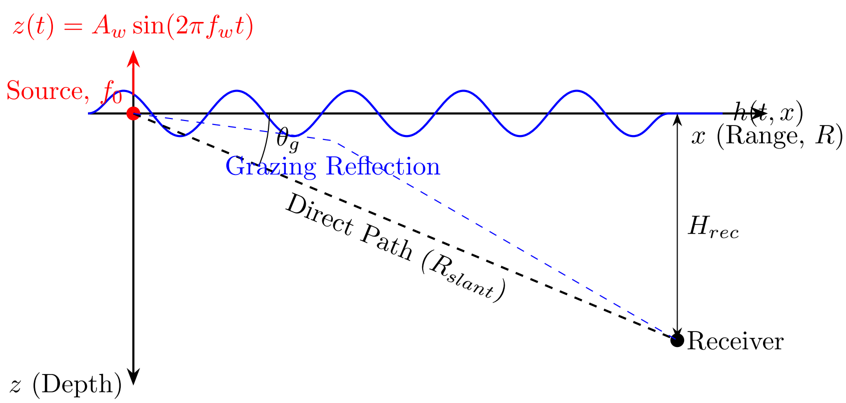
1 Theoretical Framework

This section establishes the physical foundations of acoustic signal distortion. A **Doubly Spread Channel** [1, 2] is driven by two coupled mechanisms: **Source Kinematics (Heave)** and **Dynamic Boundary Interaction (Micro-Multipath)**.

2 Source on the surface

A surface vessel represents the worst-case scenario for acoustic smearing due to physical coupling with the dynamic boundary.

2.1 Schematic Diagram



2.2 Physical & Mathematical Breakdown

2.2.1 The Grazing Angle

This subsection calculates the reflection geometry to prove the direct and surface paths overlap.

$$\theta_g = \arctan\left(\frac{H_{rec} - H_{src}}{R}\right) \tag{1}$$

For a surface source ($H_{src} \approx 0$) and long range ($R \gg H_{rec}$), $\theta_g \approx 0°$.

2.2.2 The Transmitted Signal: Source Kinematics

This models how vertical vessel heave modulates the emitted acoustic wave. The source emits a pressure wave $p(t)$:

$$p(t) = P_0 \cos(2\pi f_0 t) \tag{2}$$

The heave motion induces a phase shift:

$$z(t) = A_w \sin(2\pi f_w t) \tag{3}$$

Creating a Phase/Frequency Modulated (FM) signal:

$$p_{tx}(t) = P_0 \cos\left(2\pi f_0 t + \beta \sin(2\pi f_w t)\right) \tag{4}$$

Where modulation index β is:

$$\beta = \frac{2\pi f_0}{c} A_w \cos\theta_g \tag{5}$$

The resulting time-varying instantaneous frequency $f_i(t)$ is:

$$f_i(t) = \frac{1}{2\pi}\frac{d\Phi}{dt} = f_0 + \Delta f \cos(2\pi f_w t) \tag{6}$$

$$\Delta f = \beta f_w = \frac{2\pi f_0 f_w}{c} A_w \cos\theta_g \tag{7}$$

Spectral Expansion (The Bessel Comb)

This proves the vessel's heave expands the tonal into an infinite series of frequency sidebands [3]. Using the Jacobi-Anger expansion:

$$p_{tx}(t) = P_0 \mathrm{Re}\left\{ \sum_{n=-\infty}^{\infty} J_n(\beta) e^{i2\pi(f_0 + n f_w)t} \right\}$$

$$p_{tx}(t) = P_0 \sum_{n=-\infty}^{\infty} J_n(\beta) \cos\left(2\pi(f_0 + n f_w)t\right) \tag{8}$$

2.2.3 Channel Interaction with Dynamic Boundary & Multipath

This models how moving waves act as a dynamic scatterer (Doppler) that changes the reflected path. Because $\theta_g \approx 0°$, the time delay between paths approaches zero ($\Delta\tau \approx 0$).

$$p_{rx}(t) = \frac{1}{R_{dir}} p_{tx}(t - \tau_{dir}) + \frac{V(t)}{R_{surf}} p_{tx}(t - \tau_{surf}) \tag{9}$$

Where the time-varying scattering coefficient $V(t)$ is:

$$V(t) = -R_0 \exp\left(i\frac{4\pi f_0}{c} h_s(t) \sin\theta_g\right) \tag{10}$$

This standard Kirchhoff approximation [4, 5] encapsulates the 180° phase inversion at the boundary (-1), the amplitude loss from roughness (R_0), and the Doppler phase shift caused by the changing path length over the moving waves (exp).

2.2.4 The Smeared Received Signal

Combining source modulation and channel interaction yields the final smeared equation. Because $\tau_{dir} \approx \tau_{surf}$, we factor the received field:

$$p_{rx}(t) \propto p_{tx}(t - \tau_{dir}) \left[1 - R_0 \exp\left(i\frac{4\pi f_0}{c} h_s(t) \sin\theta_g\right)\right]$$

Conclusions:

1. **Modulated Carrier:** p_{tx} is FM due to heave.
2. **Multiplicative Fading:** The bracketed term causes random amplitude oscillation.
3. **Secondary Phase Modulation:** The exponential causes Doppler spread.

2.2.5 Connection to the Doubly Spread Channel

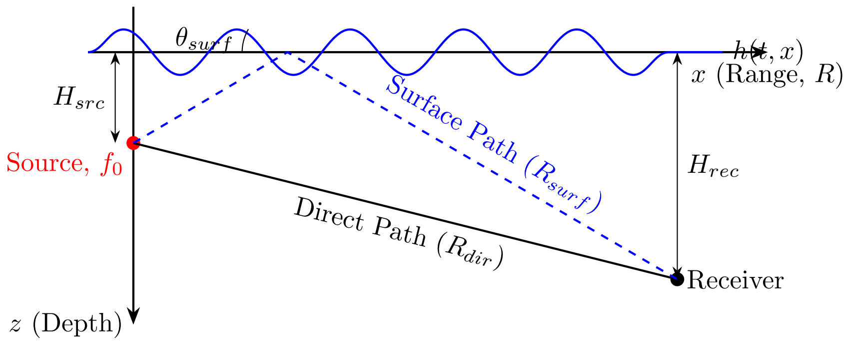
This formally defines the environment as a doubly spread acoustic channel:

1. **Delay Spread:** Paths arrive simultaneously, fading multiplicatively.
2. **Doppler Spread:** Surface velocity and vessel heave spread frequency.

3 Source Underwater

This isolates the source to demonstrate how physical decoupling stabilizes the signal.

3.1 Schematic



3.2 Physical & Mathematical Breakdown

3.2.1 Elimination of Source Kinematics

Operating below the wave base eliminates physical heave ($A_w \approx 0$).

$$\beta = \frac{2\pi f_0}{c} A_w \cos\theta_g \approx 0 \tag{11}$$

The transmitted signal reverts to a pure tone:

$$p_{tx}(t) = P_0 \cos(2\pi f_0 t) \tag{12}$$

3.2.2 Resolution of Macro-Multipath

Depth creates significant geometric path differences ($R_{surf} \gg R_{dir}$), creating macroscopic time delays ($\Delta\tau \gg 0$) that break multiplicative interference.

3.2.3 The Decoupled Received Signal

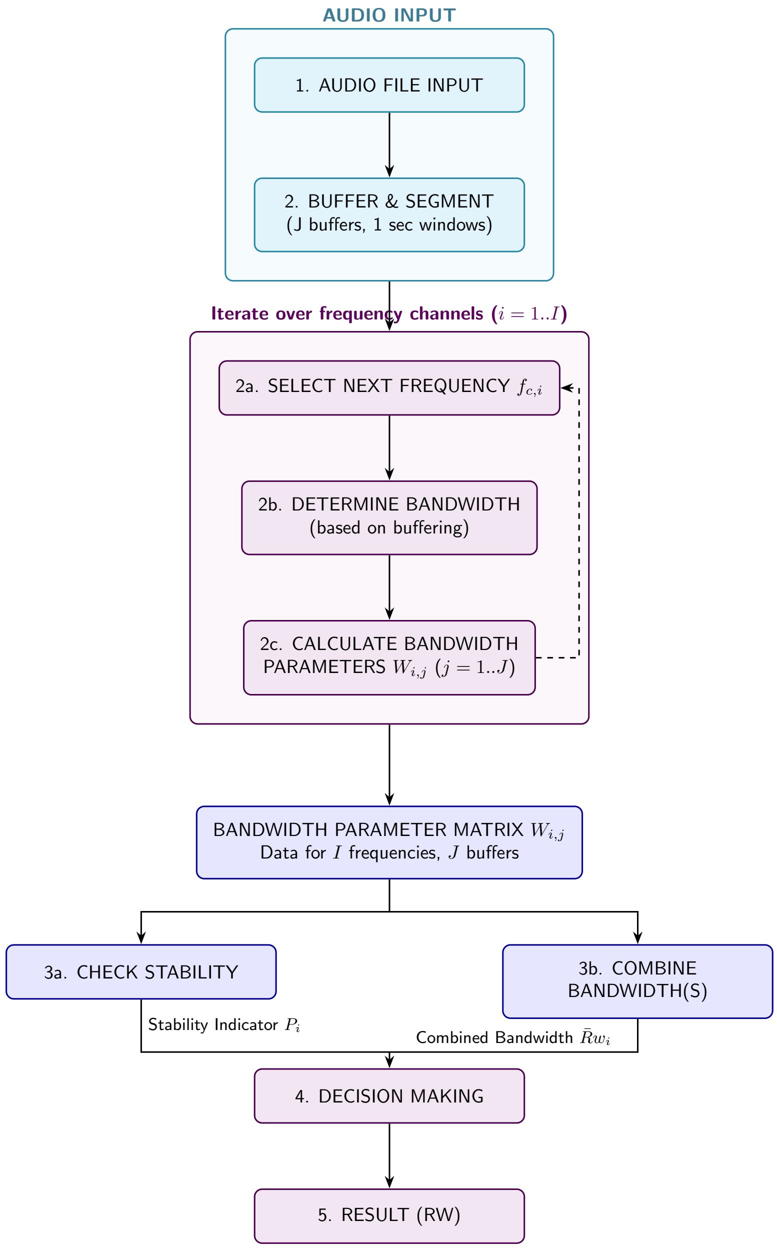
The final equation for an underwater source:

$$p_{rx}(t) = \underbrace{\frac{P_0}{R_{dir}} \cos(2\pi f_0 (t - \tau_{dir}))}_{\text{Direct Path}} + \underbrace{\frac{P_0 V(t)}{R_{surf}} \cos(2\pi f_0 (t - \tau_{surf}))}_{\text{Surface Echo}} \tag{13}$$

The direct path is unmodified, and the surface echo arrives much later, adding additive clutter rather than multiplicative fading.

4 Flow Diagram

This outlines the signal processing architecture designed to mitigate smearing.



References

[1] M. Stojanovic, "Recent advances in high-speed underwater acoustic communications," *IEEE Journal of Oceanic Engineering*, vol. 21, no. 2, pp. 125–136, 1996.

[2] J. C. Preisig, "Acoustic propagation considerations for underwater acoustic communications," *IEEE Communications Magazine*, vol. 45, no. 12, pp. 110–117, 2007.

[3] A. B. Carlson, P. B. Crilly, and J. C. Rutledge, *Communication Systems: An Introduction to Signals and Noise in Electrical Communication*, 4th ed. McGraw-Hill, 2001.

[4] F. G. Bass and I. M. Fuks, *Wave Scattering from Statistically Rough Surfaces.* Pergamon Press, Oxford, 1979.

[5] W. I. Roderick, "Doppler spectra of acoustic signals reflected from a moving rough sea surface," Ph.D. dissertation, Yale University, New Haven, CT, 1979.